E20 Mileage Drop & Fuel Cost Estimator

Internship Capstone Project Report

Option 3 — Direct PDF Upload

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Domain: Automotive / Fuel Cost Analytics (India)
Repository: github.com/UtsavUjjwal/E20-Expense-Predictor
Live app: https://e20predictor.streamlit.app/

This report documents dataset selection, cleaning, exploratory analysis, visualisation, business insights, and recommendations. Power BI dashboard work (Task 6) is omitted as agreed; an analytics snapshot page is included instead.

1. Project Overview

India is shifting petrol supply toward E20 (20% ethanol blend). Ethanol has lower energy density than petrol, so many vehicles lose mileage and owners spend more on fuel. Compatibility also varies: some engines are certified for E20, while older or non-certified cars can lose more efficiency. This project turns that problem into a practical estimator for Indian car buyers and fleet operators.

Problem statement

Drivers still plan monthly fuel budgets using claimed km/L figures and do not know how much extra they will pay after E20 blending. There is no simple public tool that combines vehicle specs, real-world tested mileage, compatibility status, and a predicted E20 drop into a rupee cost comparison.

Project objective

- Assemble a real-world Indian passenger-car dataset covering brand, engine, launch year, claimed/tested mileage, and E20 compatibility.
- Clean and inspect the data, then explore patterns that explain mileage drop.
- Train a Random Forest model (existing project pipeline) to predict E20 mileage drop %.
- Expose the result in a Streamlit app that estimates extra monthly fuel cost.
- Translate the analysis into business insights and recommendations for consumers, OEMs, and fleet managers.

Tools used

- Python, pandas, NumPy — inspection, cleaning, statistics
- Matplotlib, Seaborn — EDA charts included in this report
- scikit-learn Random Forest Regressor — drop-percentage model ($R^2 \approx 0.9986$, MAE ≈ 0.028 on the notebook test split)
- Streamlit & joblib — public estimator app
- Git & GitHub — version control and documentation

2. Dataset Description

The working file is **E20_predictions_updated.xlsx** (221 rows × 13 columns). A companion lookup file, **car_lookup.csv**, powers the app dropdowns. Records were compiled from publicly cited Indian automotive test and specification sources: ARAI, SIAM, and Autocar India real-world mileage tests. The domain is finance + automotive operations: fuel cost impact of a national fuel-policy change.

Columns and data types

Column	Data type	Role
Brand	categorical	Vehicle manufacturer
Model	categorical	Car model name
Variant	categorical	Engine / trim variant
Engine (CC)	numeric	Displacement feature
Mileage Claimed (km/L)	numeric	OEM claimed mileage
Mileage Tested (km/L)	numeric	Real-world tested mileage
Launch Year	integer	Model introduction year
Launch Year Note	text	Rare annotation (dropped)
E20 Compatible	categorical	Compatibility label
E20_Drop_%	numeric	Original drop estimate

Mileage_E20_Estimated	numeric	Original E20 mileage
E20_Drop_%_updated	numeric	Target: updated drop %
Mileage_E20_Estimated_updated	numeric	Updated E20 mileage

Brands covered: 14 manufacturers, led by Maruti (58), Hyundai (40), Tata (24), and Toyota (22). Launch years range from 2008 to 2026. E20 labels: Yes 119, No 85, CNG 7, Diesel 7, upcoming/discontinued 3.

3. Data Cleaning Process

Inspection findings

- Raw size: 221 records, 13 columns; full-row duplicates: 0.
- Engine (CC) missing on 4 rows; claimed and tested mileage missing on 13 rows each.
- E20 drop and updated drop missing on 17 rows (typically non-applicable or incomplete specs).
- Launch Year Note is empty on 219/221 rows and was dropped as non-informative.
- CNG and diesel rows are labelled N/A because ethanol blending does not apply to those fuels.
- A few luxury/high-displacement engines (for example ~3956 CC) sit far from the mass-market 1.0–1.5 L cluster.

Missing-value handling

Column	Missing values	Share of 221 rows
Brand	0	0.0%
Model	0	0.0%
Variant	0	0.0%
Engine (CC)	4	1.8%
Mileage Claimed (km/L)	13	5.9%
Mileage Tested (km/L)	13	5.9%
Launch Year	0	0.0%
Launch Year Note	219	99.1%
E20 Compatible	0	0.0%
E20_Drop_%	17	7.7%
Mileage_E20_Estimated	19	8.6%
E20_Drop_%_updated	17	7.7%
Mileage_E20_Estimated_updated	19	8.6%

Missing values were not blindly filled. Engine CC and the target drop percentage are structurally required, so those rows were excluded from statistical modelling and from the EDA analysis set. Mileage-based charts drop only the rows missing claimed/tested km/L. CNG/diesel rows are retained in the compatibility mix (to show the market) but removed from petrol-drop analysis. Categorical labels were trimmed of whitespace. Key-level duplicates on brand/model/variant/engine/year: 0 (removed if present). The cleaned analysis set used for EDA has **204 petrol-relevant rows**.

4. Exploratory Data Analysis

Descriptive statistics

Metric	Engine (CC)	Launch Year	Mileage Claimed (km/L)	Mileage Tested (km/L)	E20_Drop_%_updat

count	204.0	204.0	202.0	202.0	204.0
mean	1395.26	2019.97	19.13	15.67	13.07
std	433.69	4.09	3.72	3.27	2.98
min	998.0	2008.0	7.0	5.0	10.0
25%	1197.0	2019.0	17.35	13.82	10.0
50%	1199.0	2021.0	19.37	16.0	11.0
75%	1497.0	2023.0	20.97	17.5	16.0
max	3956.0	2026.0	27.97	24.0	18.0

Mean claimed mileage is 19.13 km/L versus mean tested mileage of 15.67 km/L — about a 18.1% real-world shortfall before E20 is applied. The updated E20 drop averages 13.07% (median 11.0%). Compatible cars average 10.66% drop versus 16.44% for non-compatible petrol cars.

Correlation and outliers

Launch year vs drop correlation is -0.71. Engine size vs drop is -0.20. Tested mileage vs drop is -0.04. IQR outlier counts in the analysis set: engine size 33, tested mileage 8, E20 drop 0. Large-displacement outliers (premium engines) were kept in descriptive EDA but are flagged in the app when engine CC is outside 600–3000 CC, because predictions there are less reliable.

Main patterns

- Among cars launched 2022 or later in the analysis set, 86% are labelled E20 compatible — OEMs have shifted quickly.
- Among cars launched 2018 or earlier, 83% are not compatible, so older fleets carry more E20 risk.
- Claimed mileage systematically overstates real-world km/L; cost planning should start from tested mileage.
- Lowest average drop brands in this sample: Lexus (10.0%), MG Motor (10.2%), Volkswagen (10.3%).
- Highest average drop brands in this sample: Tata (13.7%), Hyundai (13.7%), Maruti (14.2%).
- The original E20_Drop_% column clustered around 3–9%, while the updated target is higher (about 10–18%), which is why the app and this report use the updated field.

5. Visualizations

At least five labelled charts were produced from the cleaned dataset. Each chart is tied to a decision: market coverage, compatibility mix, physical engine spread, claimed vs real mileage, relationships among numeric fields, and drop by brand/year.

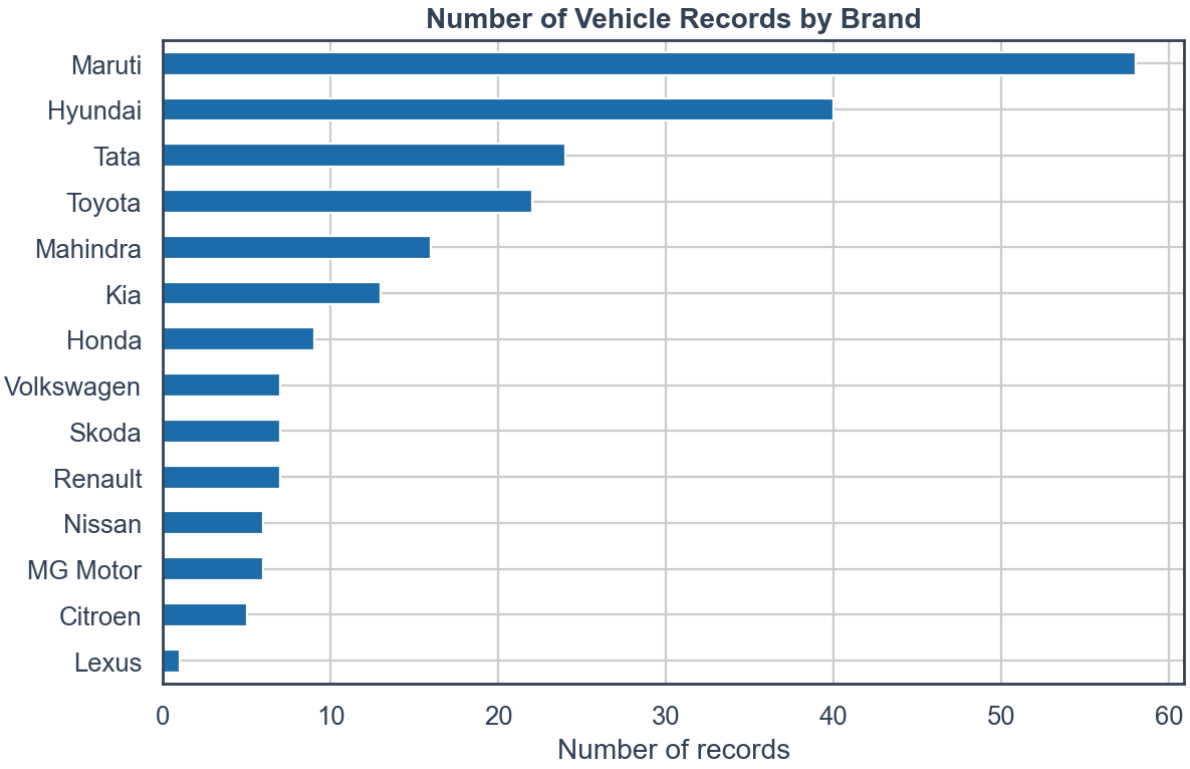


Figure 1. Records by brand — sample is India-mass-market heavy (Maruti, Hyundai, Tata).

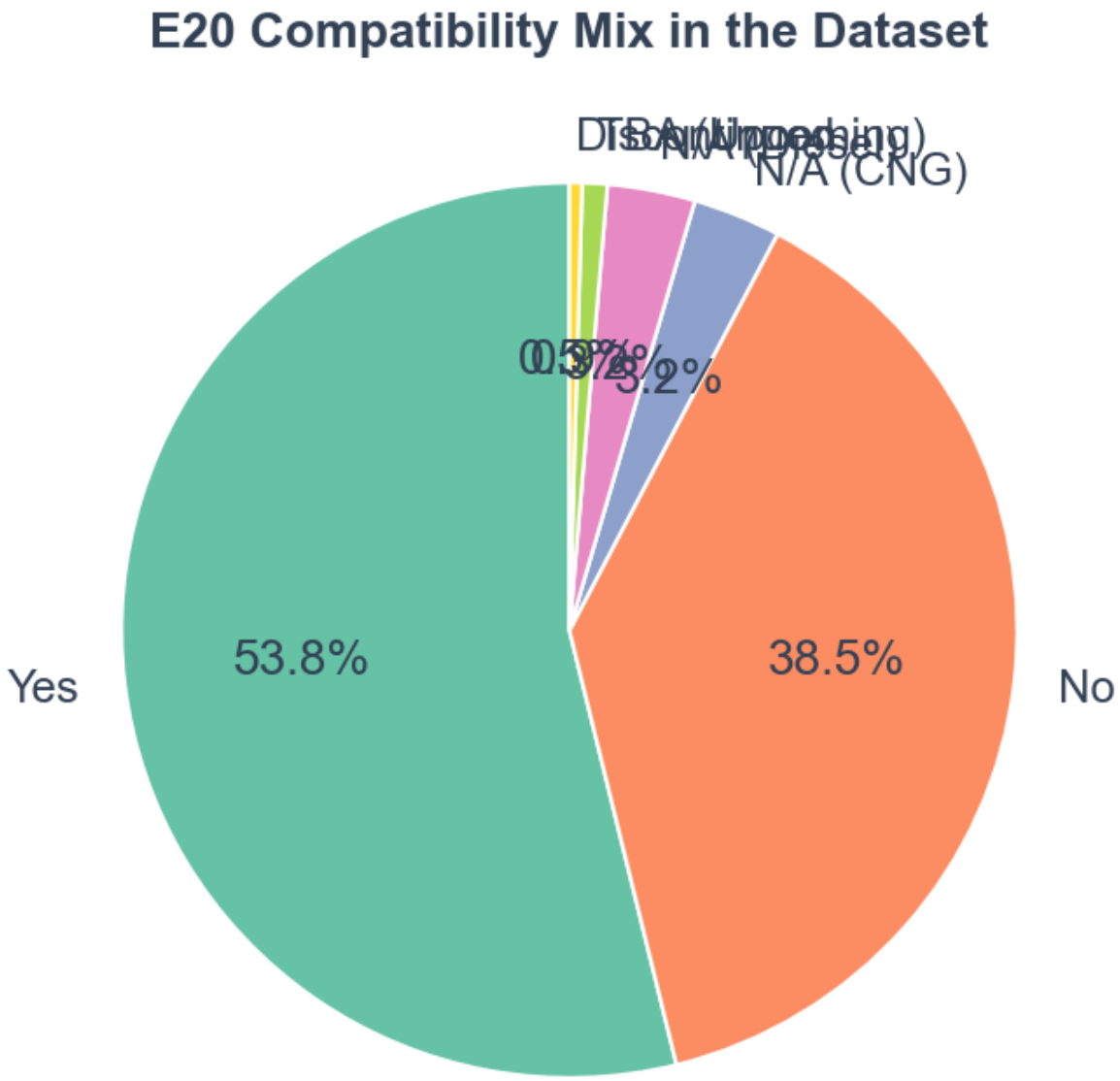


Figure 2. Compatibility mix — a large minority of cars are still not E20 certified.

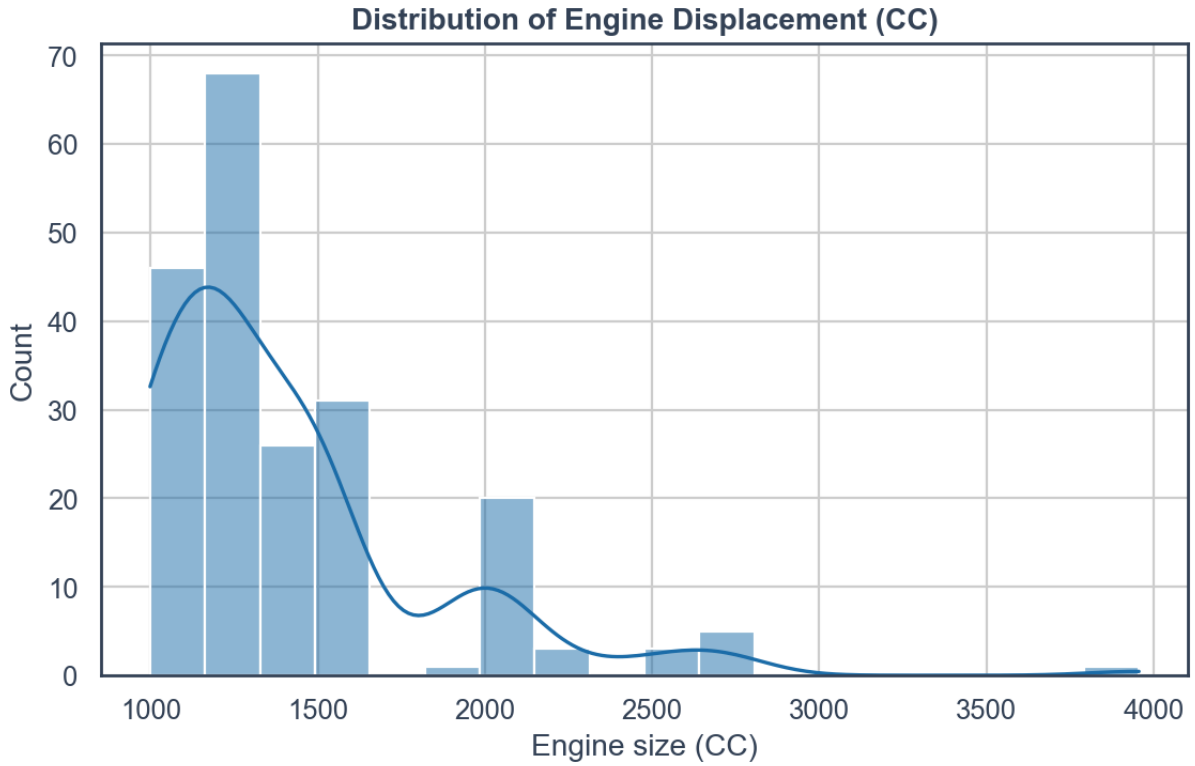


Figure 3. Engine sizes concentrate around 1.2 L and 1.5 L, the typical hatchback/SUV core.

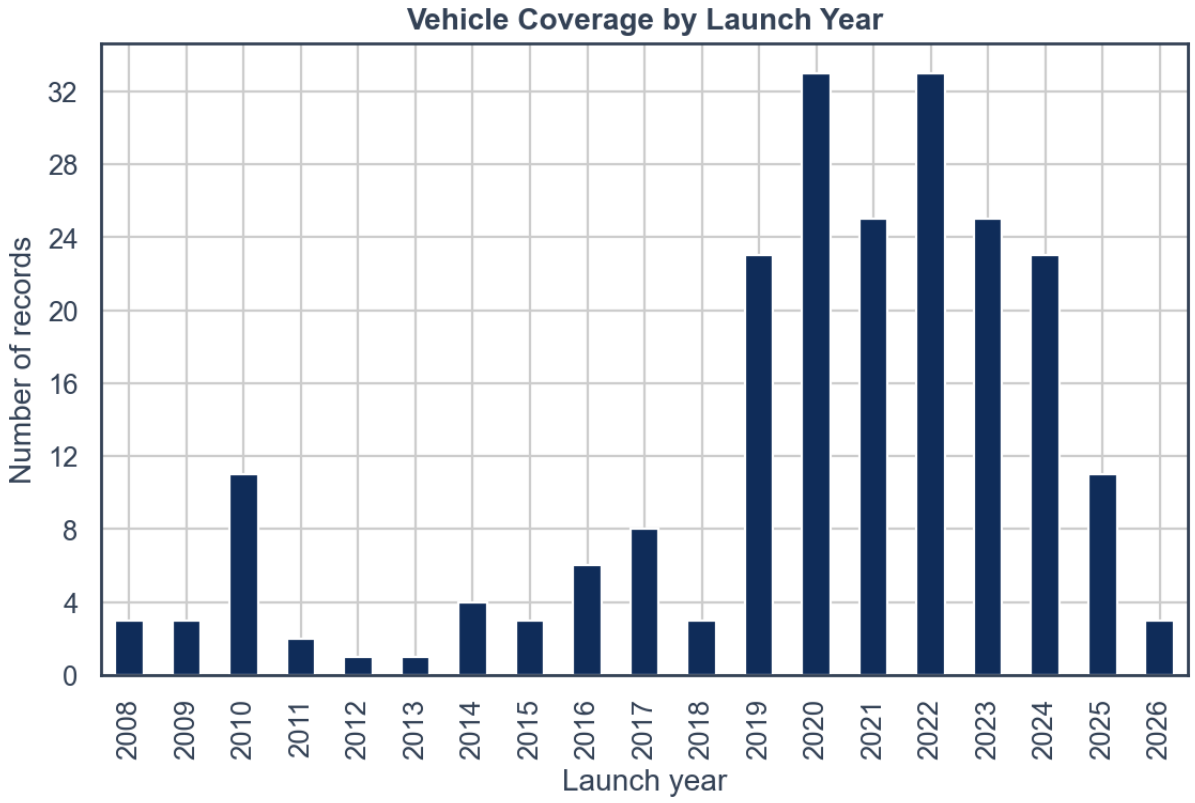


Figure 4. Most records are 2019–2024 models, matching the E20 transition window.

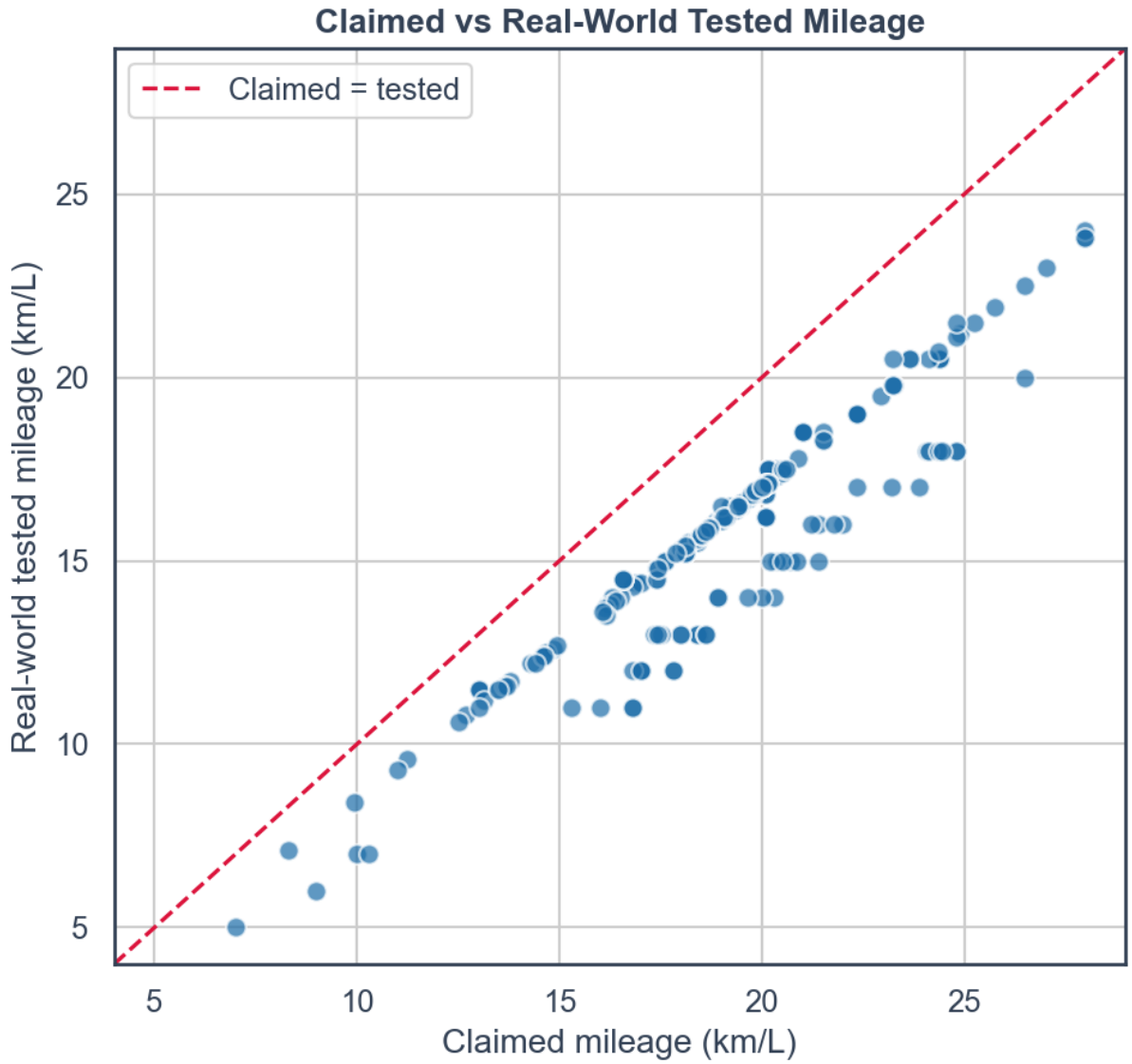


Figure 5. Almost all points sit below the equality line: tested km/L is lower than claimed.

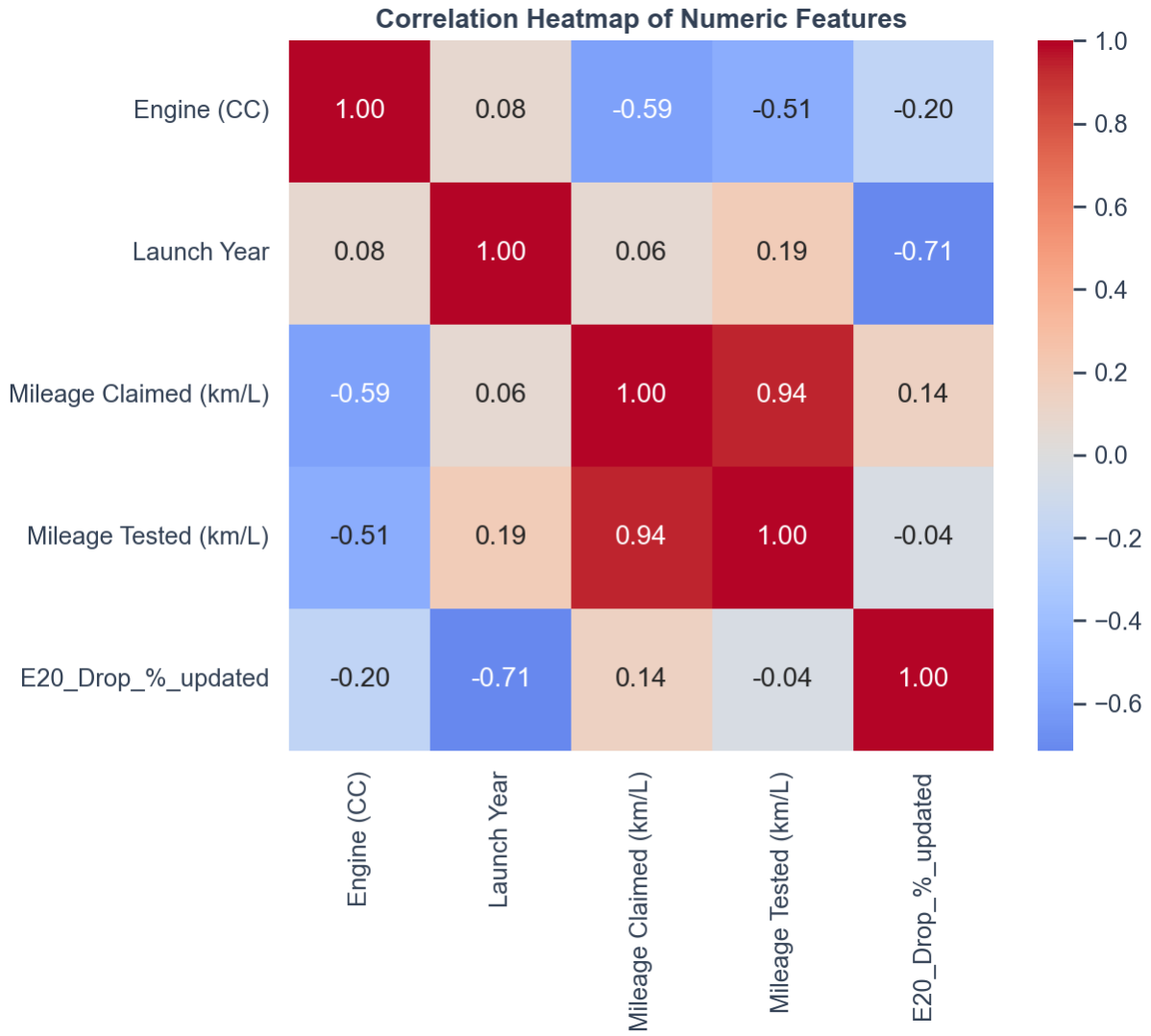


Figure 6. Correlation heatmap for engine, year, mileage, and updated drop.

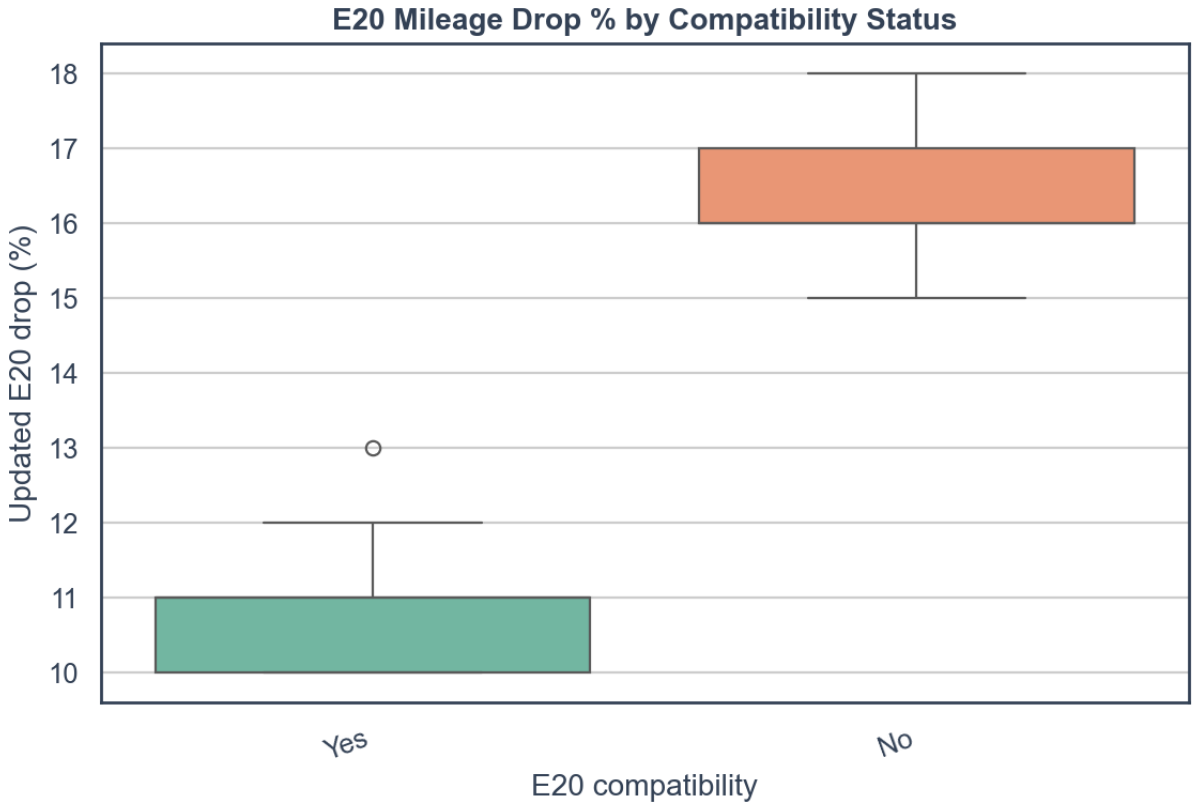


Figure 7. Drop distribution by compatibility status.

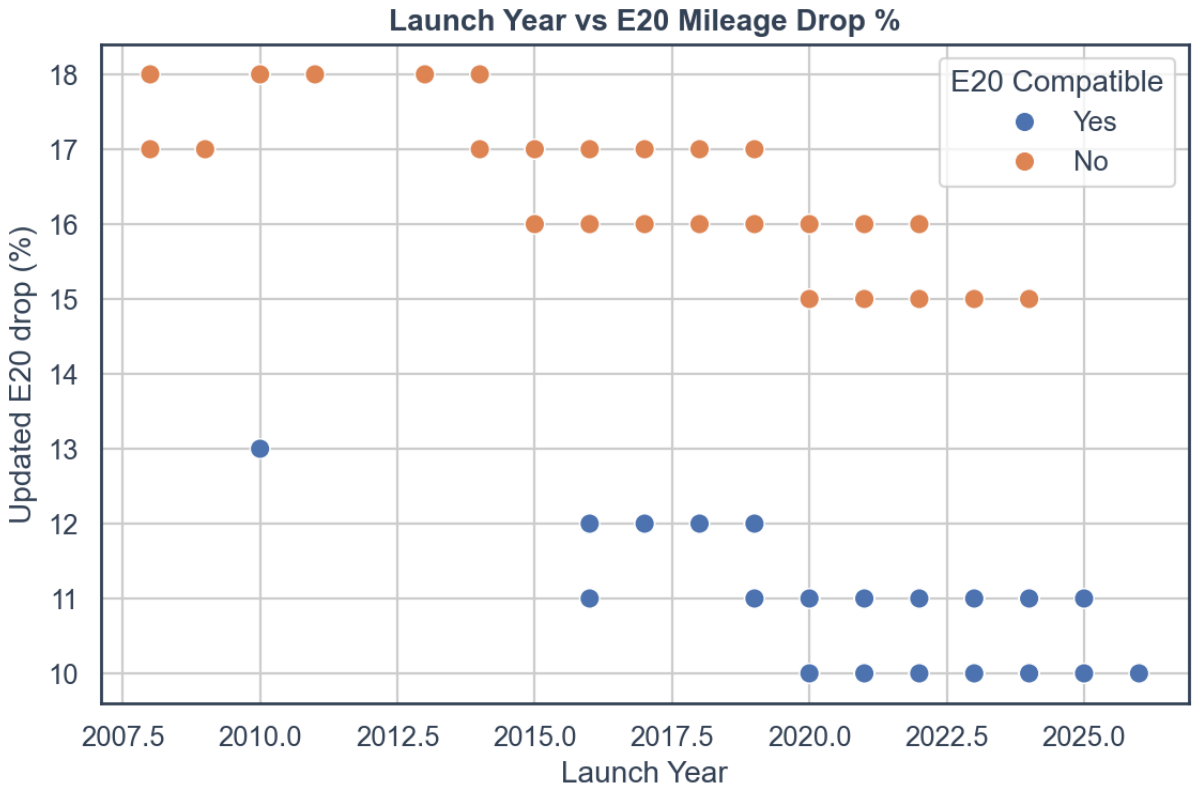


Figure 8. Newer compatible cars still show a drop, but older non-compatible cars cluster higher.

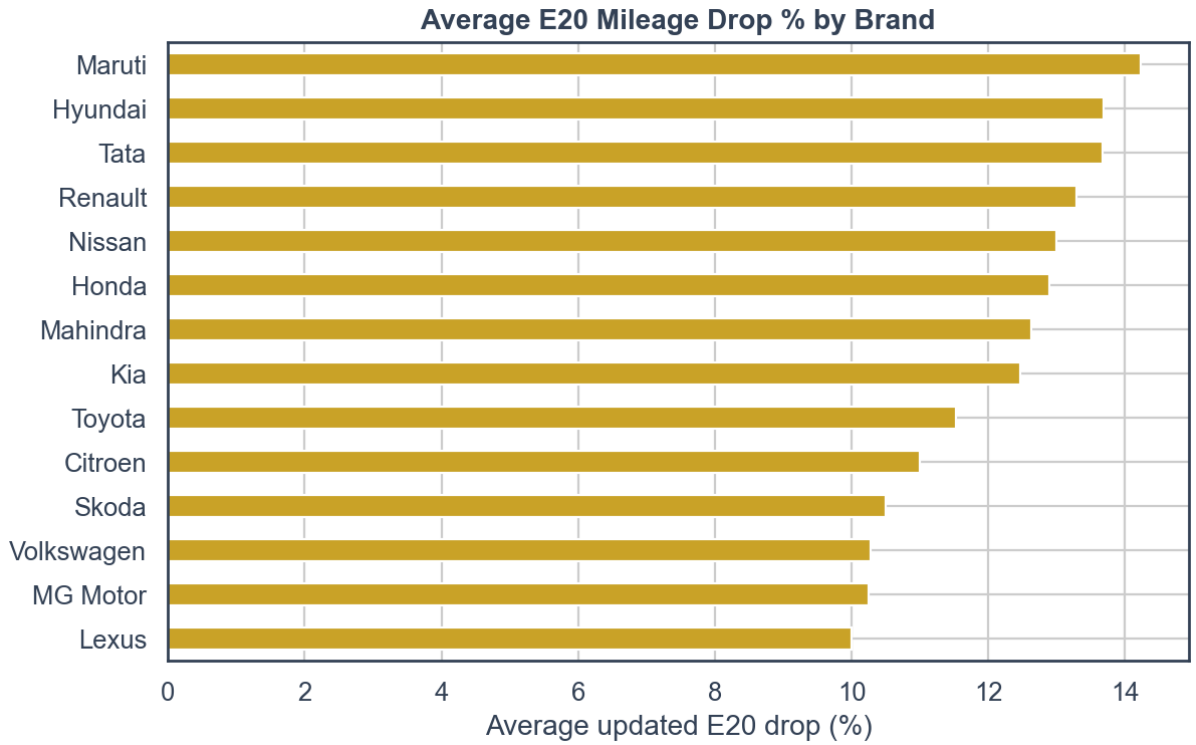


Figure 9. Brand-level average drop — useful for fleet buying shortlists.

6. Analytics Snapshot (Power BI Task Skipped)

Task 6 asked for an interactive Power BI dashboard (KPI cards, four visuals, slicers). That task is skipped per the project instruction. The snapshot below is a static stand-in: four KPI cards plus three supporting views generated from the same metrics a dashboard would highlight.

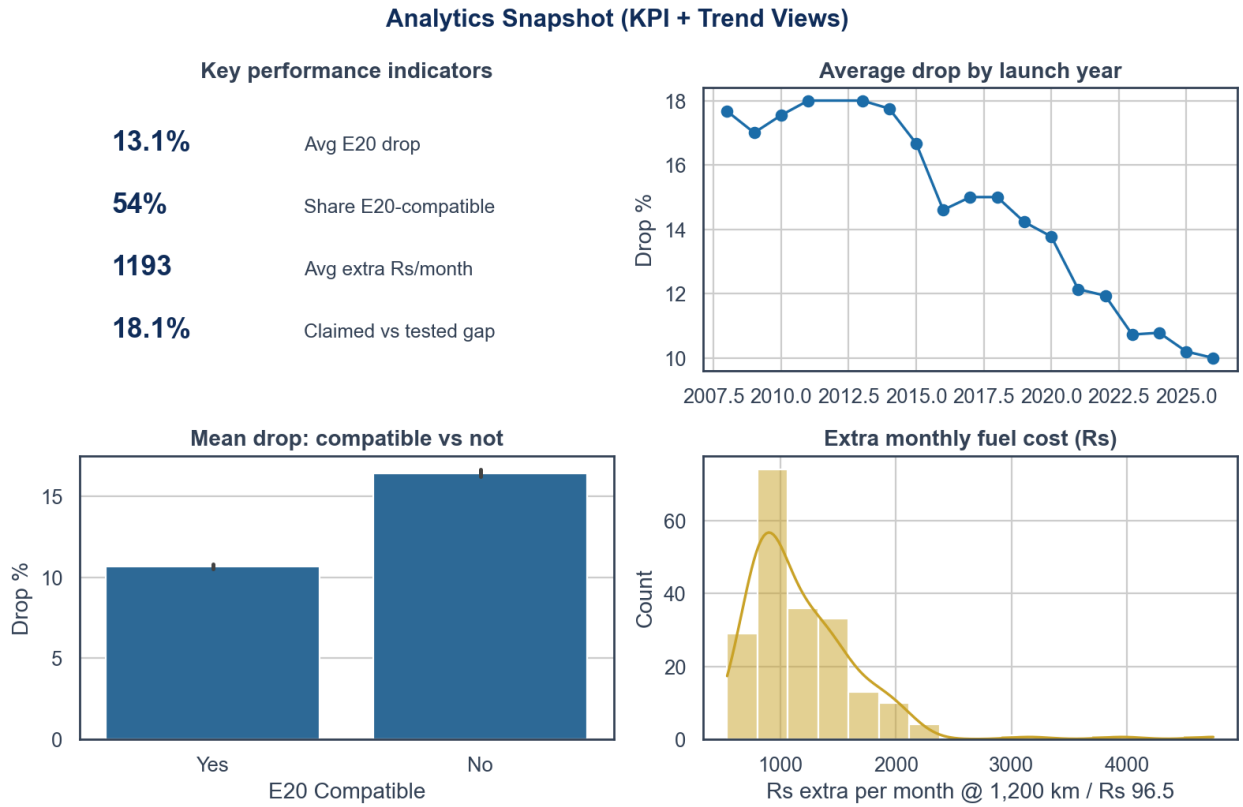


Figure 10. KPI snapshot using 1,200 km/month and Rs 96.5/litre as a standard household scenario.

- KPI 1 — Average updated E20 drop: 13.1%.
- KPI 2 — Share of records labelled E20 compatible: 54%.
- KPI 3 — Average extra monthly fuel spend in the standard scenario: Rs 1193.
- KPI 4 — Average claimed-versus-tested mileage gap: 18.1%.
- The 90th percentile extra monthly cost is Rs 1809, so high-drop / low-mileage cars are materially more expensive to run on E20.

7. Business Insights

- **Insight 1 — E20 is a real cost, not a rounding error.** Mean drop of 13.1% on already-lower tested mileage raises monthly fuel bills by about Rs 1193 for a 1,200 km driver at Rs 96.5/L.
- **Insight 2 — Certification matters.** Non-compatible petrol cars average 16.4% drop versus 10.7% for compatible cars. Buying an uncertified used car during the E20 rollout is a measurable penalty.
- **Insight 3 — Brochure mileage is the wrong planning number.** Claimed km/L overstates tested km/L by roughly 18.1%. Combining brochure figures with E20 drop would understate fuel need twice.
- **Insight 4 — The transition is front-loaded in new cars.** 86% of 2022+ analysis records are compatible, while 83% of 2018-or-earlier cars are not. Used-car and older fleet segments are the high-risk pocket.
- **Insight 5 — Brand mix is not uniform.** Average drop differs across manufacturers in this sample, so fleet RFPs can rank OEMs on expected E20 penalty rather than treating petrol as interchangeable.
- **Insight 6 — CNG/diesel are a different policy problem.** Those rows are correctly treated as 0% E20 drop. For mixed fleets, ethanol policy mainly hits petrol SKUs; CNG already decouples the household from petrol energy density.

8. Recommendations

- **Recommendation 1 (consumers & used-car platforms):** Show E20 compatibility and an estimated extra monthly rupee cost next to every petrol listing. Default the calculator to tested mileage, not claimed mileage. This is a data-driven UX change using fields already in the dataset.
- **Recommendation 2 (fleet managers / ride-hailing):** Prioritise E20-certified petrol vehicles or shift high-mileage duty cycles to CNG/hybrid. On a 1,200 km/month duty cycle the average extra is modest, but the upper tail (≈ Rs 1809/month) compounds across hundreds of cars.
- **Recommendation 3 (OEMs):** Publish certified E20 km/L, not only petrol claimed km/L, especially for 1.2 L and 1.5 L volume engines that dominate this dataset.
- **Recommendation 4 (policy / fuel retailers):** Pair E20 pump labelling with a simple household cost explainer. The analysis shows people already lose ~18% vs claimed mileage; E20 stacks another double-digit percent on top.

9. Conclusion

The dataset is small but decision-useful: 221 Indian passenger-car records spanning the E20 transition, cleaned to a petrol analysis set with explicit missing-value rules. EDA shows a consistent real-world mileage gap, a typical E20 drop near 13.1%, and a clear split between certified and uncertified cars. Visualisations communicate brand coverage, compatibility, engine mix, claimed-versus-tested km/L, correlations, and cost impact. The existing Streamlit app turns the same logic into a consumer tool. Power BI was not built for this submission. Next improvements would be a larger independent mileage-test sample, city-level fuel prices, and a dashboard with brand/year slicers.

10. Git & GitHub Workflow (Tasks 1–2 and 8)

The project lives at github.com/UtsavUjjwal/E20-Expense-Predictor. Local Git is in use; the repository was created on GitHub, cloned, and updated with multiple meaningful commits (dataset uploads, Streamlit app, EDA notebook, mileage-age adjustment, README, and this report). Standard workflow practised: clone → edit → add → commit → push, plus pull to stay in sync with origin/main.

- Task 1: Git installed, GitHub account in use, repository created, cloned locally, files added, committed, and pushed.
- Task 2: Repeated add/commit/push/pull while iterating on EDA, the predictor app, and documentation (well above three meaningful commits).
- Task 8: README and this PDF cover title, overview, problem, dataset, tools, cleaning, EDA, visuals, insights, recommendations, and conclusion.

Appendix — How to reproduce

Clone the repository, install requirements (pandas, matplotlib, seaborn, openpyxl, reportlab, scikit-learn, streamlit, joblib), then run **python generate_capstone_report.py** to rebuild this PDF and the figure folder. Run **streamlit run app.py** for the live estimator.